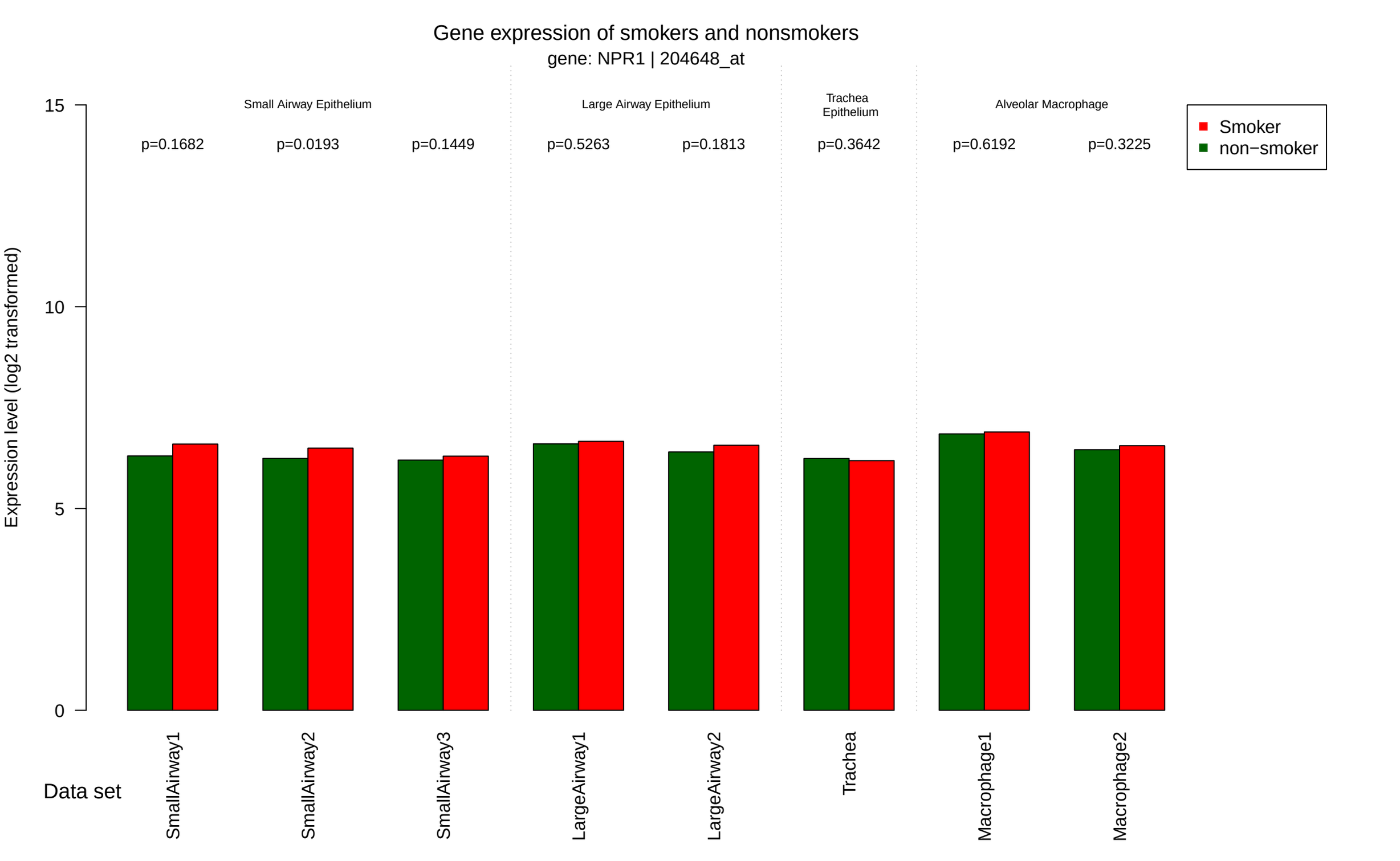
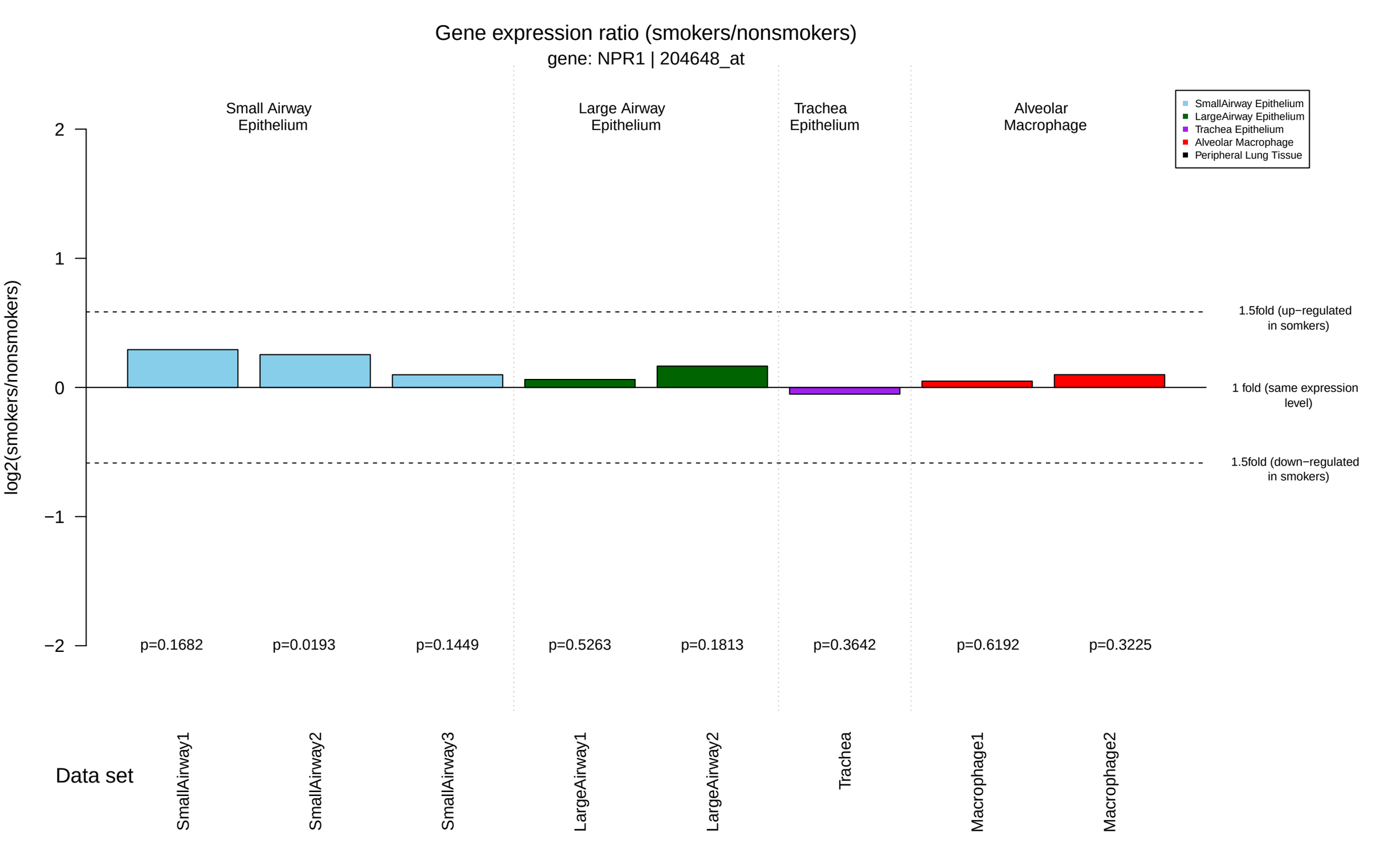
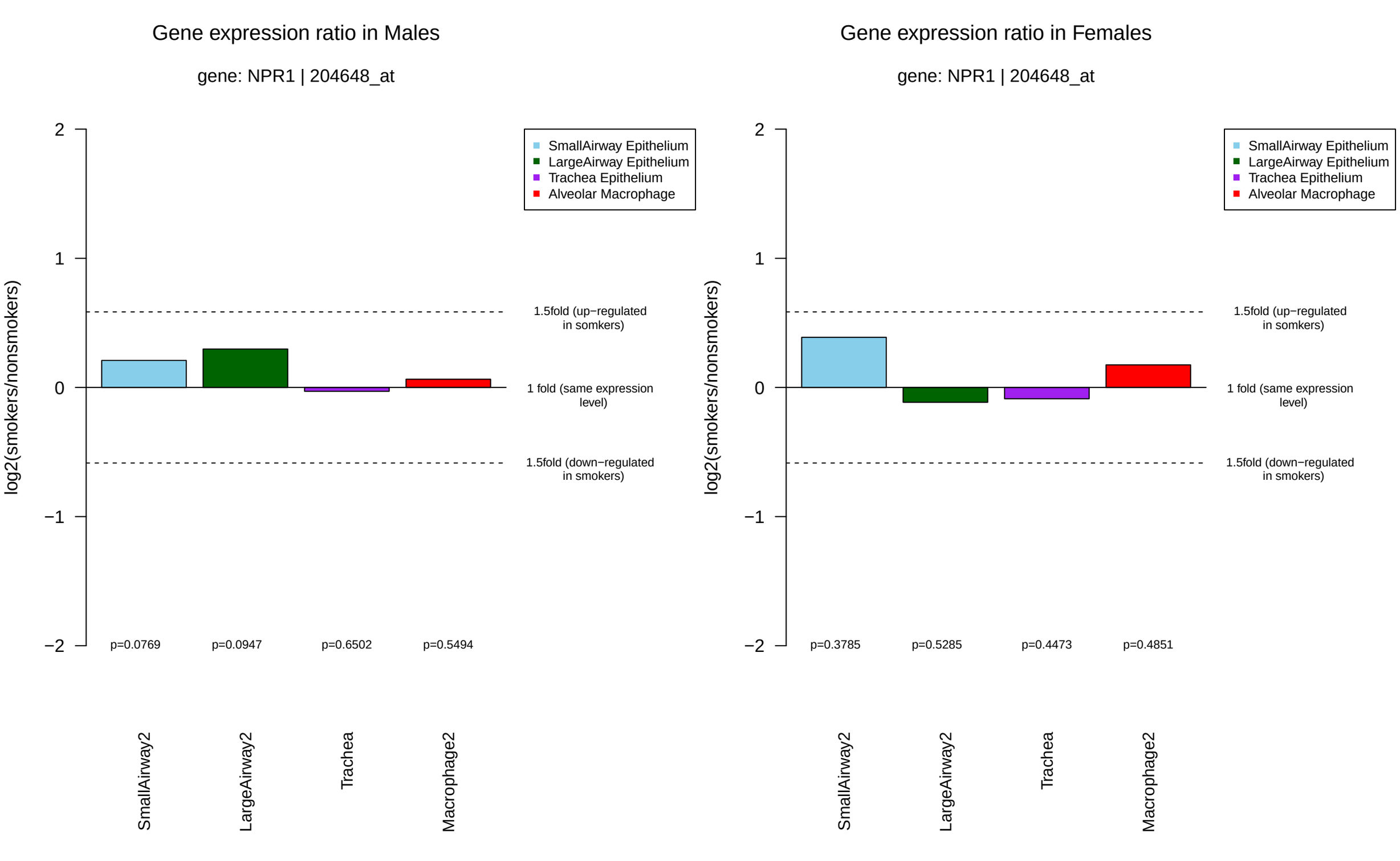
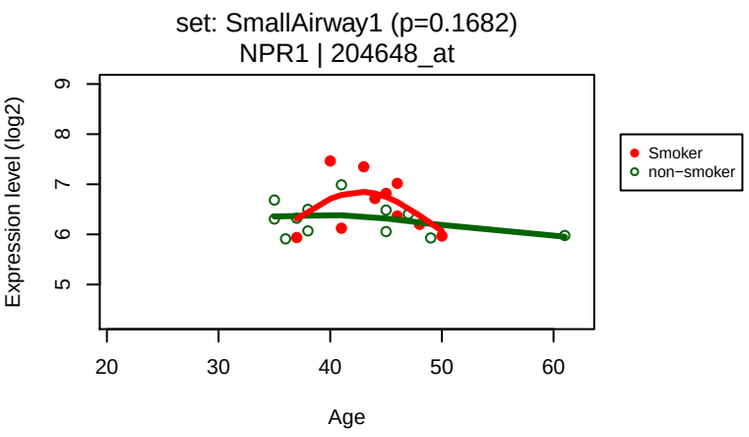
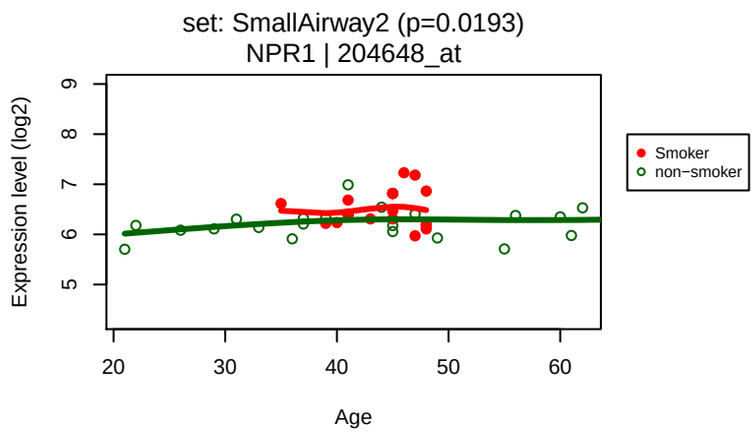
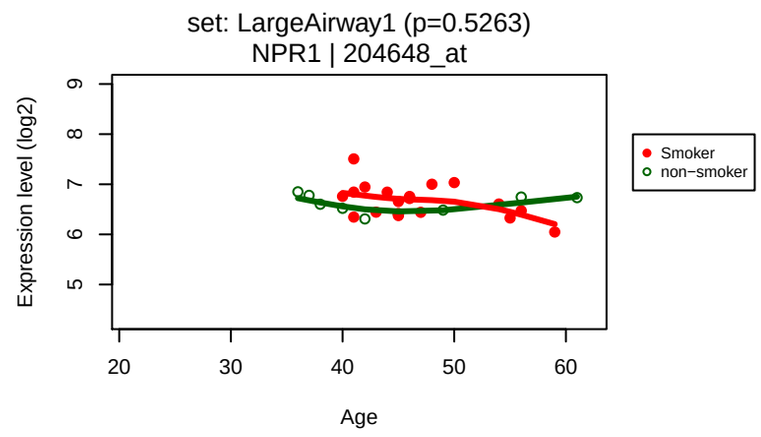

Gene expression of smokers and nonsmokers
gene: NPR1 | 204648_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1682
p=0.0193
p=0.1449
p=0.5263
p=0.1813
p=0.3642
p=0.6192
p=0.3225
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

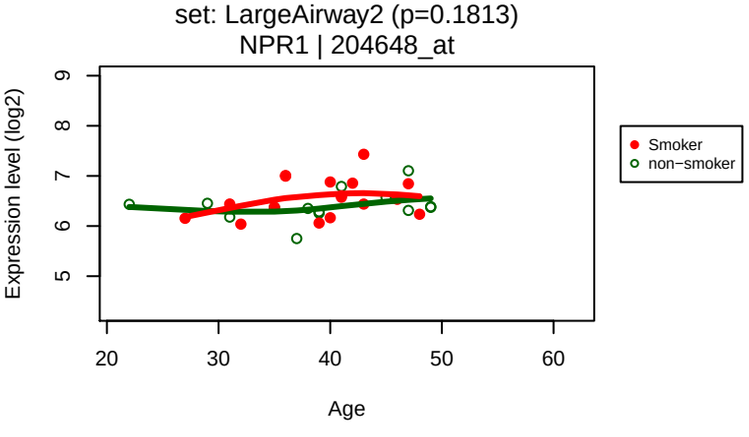

Gene expression ratio (smokers/nonsmokers)
gene: NPR1 | 204648_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1682
p=0.0193
p=0.1449
p=0.5263
p=0.1813
p=0.3642
p=0.6192
p=0.3225
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

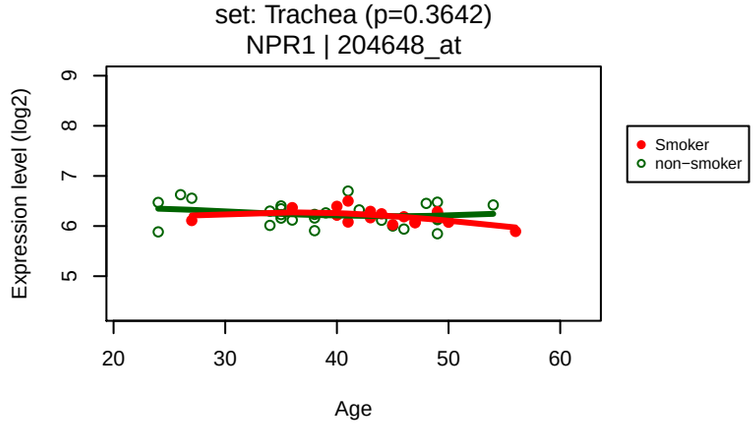

Gene expression ratio in Males
gene: NPR1 | 204648_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0769
p=0.0947
p=0.6502
p=0.5494
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPR1 | 204648_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3785
p=0.5285
p=0.4473
p=0.4851
SmallAirway2
LargeAirway2
Trachea
Macrophage2

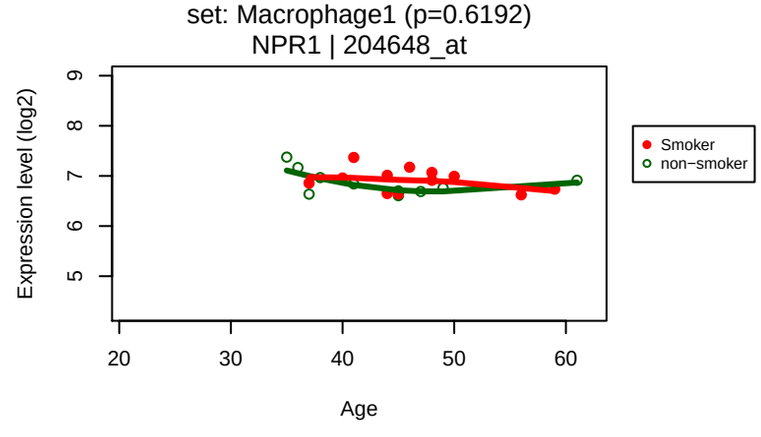

set: SmallAirway1 (p=0.1682)
NPR1 | 204648_at
Expression level (log2)
Age
Smoker
non−smoker

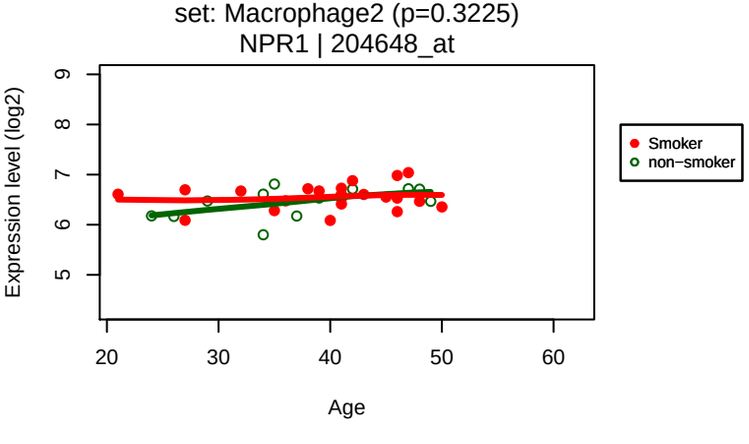

set: SmallAirway2 (p=0.0193)
NPR1 | 204648_at
Expression level (log2)
Age
Smoker
non−smoker

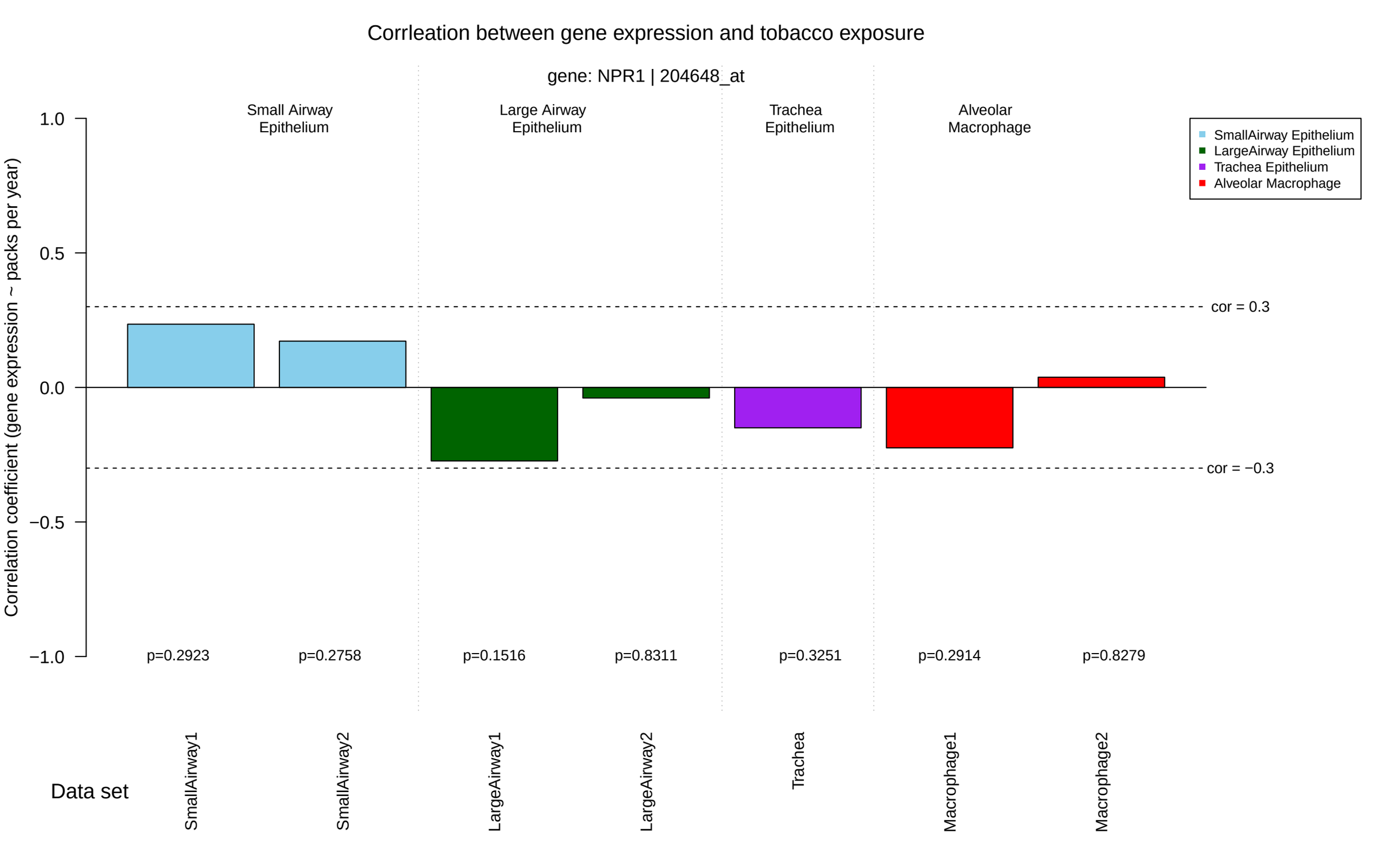

set: LargeAirway1 (p=0.5263)
NPR1 | 204648_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1813)
NPR1 | 204648_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3642)
NPR1 | 204648_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6192)
NPR1 | 204648_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3225)
NPR1 | 204648_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPR1 | 204648_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2923
p=0.2758
p=0.1516
p=0.8311
p=0.3251
p=0.2914
p=0.8279
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2